Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Simplification of algebraic expressions: order of operations, brackets and word problems. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Second Term
Week	9
Topic	Simplification of Algebraic Terms
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems
Theme	—
Curriculum objectives	• carry out order of operation in algebra • insert/remove bracket and simplify expression • solve word problems involving algebraic expression
Previous knowledge	• Addition and subtraction of algebraic terms • Multiplication and division of algebraic terms
Lesson sequence	• carry out order of operation in algebra — the teacher solves some questions in order of operation on the board; the student listen, contribute, and where necessary ask questions where they do not understand; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$ • insert/remove bracket and simplify expression — the teacher solves some questions on insertion, removal of bracket on the board; the student listen, ask questions on their areas of difficulty; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, marks their notes and makes corrections. Set for the students: (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$ • solve word problems involving algebraic expression — the teacher solves some questions on word problems involving algebraic expression on the board; the student listen, ask questions where they do not understand; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. A pen cost h naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.
Instructional materials	• Chart showing algebraic expression.
Evaluation	• Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$ • Simplify $7 \times 3a - (3a + 5a) \times 2$

	• Simplify $12-(7-2f)-(3-f)$ • Simplify $(8a-5)+(14-3a)$ • A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils.

Lesson objectives

1.	carry out order of operation in algebra
2.	insert/remove bracket and simplify expression
3.	solve word problems involving algebraic expression

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Simplification of algebraic expressions: order of operations, brackets and word problems

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: The Right Order: Working Through Algebraic Expressions

An algebraic expression is a sentence of mathematics built from letters (like x and y) and numbers, joined by operations such as +, −, × and ÷. For example, 2x + 3 and 4(x − 1) + 5 are both algebraic expressions.

When an expression has more than one operation, you cannot simply work from left to right. Everyone must get the same answer, so mathematicians agree on one fixed order to follow. The order is often remembered with the word BODMAS:

- Brackets — work out anything inside brackets first
- Orders — powers, such as x^2, next
- Division and Multiplication — from left to right
- Addition and Subtraction — from left to right, last

The same order is used in algebra as with plain numbers. The letters simply stand in for numbers, so they follow exactly the same rules. Whenever you see brackets, deal with them first — and remember that a number written next to a bracket, like 3(x + 2), means multiply everything inside by that number (this is called expanding the bracket).

Example 1 — Multiplication before addition
Simplify 2 + 3 × 4x.

There is no bracket and no power here. BODMAS says multiplication comes before addition, so do the 3 × 4x first.
3 × 4x = 12x.
Now add: 2 + 12x.
So 2 + 3 × 4x = 2 + 12x. We cannot add the 2 and the 12x together because one is a plain number and the other has an x, so the expression is now as simple as it can be.

Example 2 — Removing a bracket
Simplify 4(2x − 3).

The bracket is first, and the 4 outside multiplies both terms inside.
4 × 2x = 8x, and 4 × (−3) = −12.
So 4(2x − 3) = 8x − 12.

Example 3 — Brackets then collecting like terms
Simplify 3x + 2(x + 5) − 7.

Step 1 — the bracket: 2(x + 5) = 2x + 10.
The expression becomes 3x + 2x + 10 − 7.
Step 2 — collect the x-terms: 3x + 2x = 5x.
Step 3 — collect the numbers: 10 − 7 = 3.
So 3x + 2(x + 5) − 7 = 5x + 3.

The order of operations also helps with word problems. A word problem describes a situation in sentences, and you turn it into an algebraic expression using a letter for the unknown. Read carefully, decide which operation each phrase points to (for example, "more than" means add, "times" means multiply, "the difference" means subtract), then simplify using BODMAS.

Here is a useful way to think about it: if a problem mentions a total built from separate parts, brackets often hold one part so it is dealt with before the rest of the expression.

Example 4 — A rectangle
A rectangle has length (x + 4) cm and width 3 cm. Write a simplified expression for its perimeter.

The perimeter of a rectangle is twice the length plus twice the width:
Perimeter = 2(length + width) = 2((x + 4) + 3).
Work inside the bracket first: (x + 4) + 3 = x + 7.
Now expand: 2(x + 7) = 2x + 14.
So the perimeter is 2x + 14 cm.

Example 5 — A shopping problem
A pencil costs x naira and a pen costs 5 naira more. Amina buys 3 pencils and 2 pens. Write a simplified expression for the total cost.

A pen costs x + 5 naira.
3 pencils cost 3 × x = 3x naira.
2 pens cost 2 × (x + 5) = 2x + 10 naira.
Total = 3x + 2x + 10 = 5x + 10 naira.
So the total cost is 5x + 10 naira.

Practice

1. Simplify 6 + 2 × 3x. Remember which operation comes first.
2. Simplify 4(2x − 3) + 5x.
3. Simplify 3x + 2(x + 5) − 7.
4. A rectangle has length (2x + 1) cm and width 4 cm. Write a simplified expression for its perimeter.
5. A box weighs x kg. A crate weighs 3 times as much as the box plus 2 kg more. Write a simplified expression for the weight of the crate.

Section 2: Removing and Inserting Brackets

Brackets hold terms together, and often an expression is easier to work with once you have either taken the brackets away or grouped terms inside a new pair of brackets. Both moves follow the same simple idea: whatever sits right in front of a bracket multiplies every single term inside it.

Removing a bracket
When a letter or number stands in front of a bracket, multiply it by each term inside, then drop the brackets:
- 3(x + 2) = 3·x + 3·2 = 3x + 6
- 2(b − 5) = 2·b − 2·5 = 2b − 10

When a minus sign stands in front of a bracket, you change the sign of every term inside and then drop the brackets. Think of the minus as −1 multiplying each term:
- −(a + 4) = −a − 4
- −(2y − 3) = −2y + 3   (the −3 becomes +3)

Inserting a bracket
To put terms back into a bracket, find the largest factor that divides every term, write it in front of the bracket, and divide each term by it:
- 6x + 9: 3 divides both terms, so 6x + 9 = 3(2x + 3).
- 8p − 4: 4 divides both terms, so 8p − 4 = 4(2p − 1).

Simplifying after removing brackets
Once the brackets are gone, finish by collecting the like terms — add or subtract the terms that have the same letter (and the plain numbers on their own). For example, 3x + 6 + 4x − 4 becomes (3x + 4x) + (6 − 4) = 7x + 2. Work carefully, one sign at a time, and you will always end with a tidy expression.

Example 1: Remove the brackets and simplify 4(x + 3) + 2(x − 1)

Remove each bracket first, multiplying the number in front by every term inside:
4(x + 3) = 4x + 12   and   2(x − 1) = 2x − 2

So the expression becomes: 4x + 12 + 2x − 2

Collect the like terms. The x-terms: 4x + 2x = 6x. The plain numbers: 12 − 2 = 10.

Answer: 6x + 10

Example 2: Remove the brackets and simplify 5a − (2a + 3)

The bracket is preceded by a minus sign, so change the sign of every term inside and drop the brackets: −(2a + 3) = −2a − 3

The expression becomes: 5a − 2a − 3

Collect the like terms: 5a − 2a = 3a. The number −3 has no a to join, so it stays.

Answer: 3a − 3

Example 3: Insert a bracket to write 12m − 8 as a product

Find the largest factor of both terms. 12m and 8 are both divisible by 4, so bring 4 out in front: 12m ÷ 4 = 3m   and   8 ÷ 4 = 2

Answer: 4(3m − 2). Check by removing the bracket: 4·3m − 4·2 = 12m − 8, which matches.

Practice

1. Remove the brackets and simplify: 4(x + 3) + 2(x − 1)
2. Remove the brackets and simplify: 7a − (3a + 2)
3. Remove the brackets and simplify: 2(3b + 1) − 3(b − 4)
4. Insert a bracket to write 8x − 12 as a product.
5. A rectangle has length (2x + 3) cm and width 4 cm. The perimeter is found by removing the brackets in 2(2x + 3) + 2(4). Simplify this expression.

Section 3: From Words to Algebra

A word problem gives you a story, and your job is to turn the story into algebra, work with the expression you build, and then answer the question the story is asking. You already know how to simplify algebraic terms by collecting like terms and applying the order of operations. Word problems simply ask you to spot which operation (add, subtract, multiply, divide) the words are describing, and to use a letter for the unknown amount.

The key moves are these:
1. Read the problem and decide what the unknown is. Give it a letter, such as x or y.
2. Write each sentence of the story as part of an algebraic expression.
3. Simplify the expression by collecting like terms.
4. Use the final answer to answer the original question.

Some phrases give you strong clues:
• "more than", "add", "sum of" → addition
• "less than", "difference", "take away" → subtraction
• "times", "product of", "twice", "three times" → multiplication
• "shared equally", "divided between" → division

Example 1 — One unknown.
A number x is increased by 5, and the result is then tripled. Write an expression for the final amount.

Step 1: The unknown is the number x.
Step 2: "increased by 5" means x + 5.
Step 3: "tripled" means multiply by 3, so put the whole thing in brackets: 3(x + 5).
Step 4: Expand the bracket: 3x + 15.

So the final expression is 3x + 15.

Example 2 — Two unknowns in one story.
Amina has y oranges. Her brother has 3 more than twice as many oranges as Amina. Write, in terms of y, how many oranges her brother has.

Step 1: Amina has y oranges.
Step 2: "twice as many as Amina" means 2y.
Step 3: "3 more than" that means add 3: 2y + 3.

So her brother has 2y + 3 oranges.

Example 3 — A word problem you simplify then evaluate.
A rectangle has length x cm. Its width is 2 cm less than its length.
(a) Write an expression for the perimeter of the rectangle.
(b) Find the perimeter when x = 6.

Step 1: Length = x cm. Width = 2 cm less than length, so width = x − 2.
Step 2: Perimeter = length + width + length + width = 2 × length + 2 × width.
So perimeter = 2x + 2(x − 2).
Step 3: Expand the bracket: 2x + 2x − 4 = 4x − 4.
Step 4: When x = 6, perimeter = 4(6) − 4 = 24 − 4 = 20 cm.

So (a) the perimeter is 4x − 4 cm, and (b) it is 20 cm when x = 6.

Notice how a single word problem can combine everything: building the expression from the words, then simplifying it by expanding brackets and collecting like terms, and sometimes putting in a value for the letter at the end. Keep your steps in the same order every time, and check that your final expression actually answers the question asked — not just the algebra you did.

Practice

1. A number n is decreased by 7, and the result is then multiplied by 4. Write an expression for the final amount.
2. Bala is m years old. His sister is 4 years younger than twice his age. Write, in terms of m, how old his sister is.
3. The length of a rectangle is y cm and its width is 3 cm more than the length.
   (a) Write an expression for the perimeter of the rectangle.
   (b) Find the perimeter when y = 5.

Check your work

Attempt every question before you read this page.

The Right Order: Working Through Algebraic Expressions

1. 6 + 6x

   Multiplication comes before addition. 2 × 3x = 6x, then add the 6: 6 + 6x.
2. 13x − 12

   Expand the bracket first: 4(2x − 3) = 8x − 12. The expression becomes 8x − 12 + 5x. Collect like terms: 8x + 5x = 13x. So the answer is 13x − 12.
3. 5x + 3

   Bracket first: 2(x + 5) = 2x + 10. The expression becomes 3x + 2x + 10 − 7. Collect x-terms: 3x + 2x = 5x. Collect numbers: 10 − 7 = 3. Answer: 5x + 3.
4. 4x + 10 cm

   Perimeter = 2(length + width) = 2((2x + 1) + 4). Work inside the bracket: (2x + 1) + 4 = 2x + 5. Expand: 2(2x + 5) = 4x + 10. So the perimeter is 4x + 10 cm.
5. 3x + 2 kg

   Three times the box is 3 × x = 3x kg, and 2 kg more is added last. So the crate weighs 3x + 2 kg.

Removing and Inserting Brackets

1. 6x + 10

   4(x + 3) = 4x + 12 and 2(x − 1) = 2x − 2. The expression is 4x + 12 + 2x − 2. Collect like terms: 4x + 2x = 6x and 12 − 2 = 10, so the answer is 6x + 10.
2. 4a − 2

   The minus in front of the bracket changes every sign inside: −(3a + 2) = −3a − 2. So 7a − 3a − 2 = 4a − 2.
3. 3b + 14

$2(3b + 1) = 6b + 2$ and $-3(b - 4) = -3b + 12$. The expression is $6b + 2 - 3b + 12$. Collect like terms: $6b - 3b = 3b$ and $2 + 12 = 14$, so the answer is $3b + 14$.

4. $4(2x - 3)$

   The largest factor of $8x$ and 12 is 4. Bring 4 out: $8x \div 4 = 2x$ and $12 \div 4 = 3$, so $8x - 12 = 4(2x - 3)$. Check: $4 \cdot 2x - 4 \cdot 3 = 8x - 12$.

5. $4x + 14$

   Remove the brackets: $2(2x + 3) = 4x + 6$ and $2(4) = 8$. The expression is $4x + 6 + 8$. Collect like terms: $4x$ stays, and $6 + 8 = 14$, so the perimeter is $4x + 14$.

From Words to Algebra

1. $4(n - 7) = 4n - 28$

   Decreased by 7 means $n - 7$. Multiply the whole result by 4, so bracket it: $4(n - 7)$. Expand: $4n - 28$.

2. $2m - 4$ years old

   Twice Bala's age is $2m$. Four years younger means subtract 4: $2m - 4$.

3. (a) $4y + 6$ cm; (b) 26 cm

   Length = y, width = $y + 3$. Perimeter = $2y + 2(y + 3) = 2y + 2y + 6 = 4y + 6$. When $y = 5$: $4(5) + 6 = 20 + 6 = 26$ cm.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-490b5b3c	**Date**	
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____